DERSU ÇELİKSÖZ

Founder | Senior Systems and Control Engineer

Ankara, Turkey | +90 505 610 74 79 | dersuceliksoz@gmail.com | linkedin.com/in/dersuceliksoz | dersuceliksoz.com

PROFESSIONAL SUMMARY

Senior Systems and Control Engineer with 5+ years of experience in hybrid electric vehicle control, system modeling, real-time simulation, HIL validation, and rapid control prototyping for defense/automotive applications. Currently providing independent engineering consultancy and project-based engineering services in heavy-duty vehicle electrification, model-based control, reduced-order modeling, and hybrid electric system architecture.

WORK EXPERIENCE

Founder | Senior Systems and Control Engineer

DC Engineering Solutions, Ankara, Turkey **05.2026 – Present**

- Provide system-level engineering consultancy for heavy-duty truck electrification and hybridization
- Design and deliver model-based control systems for hybrid electric vehicle projects, including regenerative braking, motion/energy management, gearbox control, and embedded implementation
- Develop reduced-order models through physics-based and data-driven methods for soft-sensing applications

Senior Controls Design Engineer

FNSS Defense Systems Inc., Ankara, Turkey **09.2021 – 05.2026**

- Architected model-based control systems for hybrid electric tracked vehicles, including motion control and power management using MATLAB/Simulink
- Developed vehicle and fault management state machines, and designed CAN-based communication architectures for system-level control and diagnostics
- Modeled hybrid powertrain and full-vehicle dynamics for control-oriented simulation and design decisions
- Developed real-time driver-in-the-loop simulators and accelerated HIL simulations using data-driven reduced-order models, achieving up to ~100x speed-up in nonlinear subsystems
- Designed and executed MIL, SIL, and HIL validation workflows to ensure system safety and reliability
- Performed rapid control prototyping using NI VeriStand/LabVIEW and code generation via Simulink Coder
- Contributed to driveline architecture development for electrified tracked vehicle platforms

Undergraduate Engineer

STM Defense Technologies and Engineering Inc., Ankara, Turkey **10.2020 – 09.2021**

- Designed and integrated UAV gimbal systems combining electromechanical design and control systems
- Modeled and optimized vibration isolation systems to improve stability of UAV subsystems

EDUCATION

Master of Science | Mechanical Engineering — CGPA: 3.64 / 4.00

Middle East Technical University (METU), Ankara, Turkey **02.2022 – 01.2025**

Curriculum focused on dynamics and control; thesis on nonlinear gear dynamics

Bachelor of Science | Mechanical Engineering — CGPA: 3.77 / 4.00

Middle East Technical University (METU), Ankara, Turkey **08.2016 – 07.2021**

SKILLS

- **Control algorithm development:** MATLAB, Simulink, Stateflow, PID
- **Modeling & real-time simulation:** Simulink, Simscape (Mechanical/Electrical/Multibody), Reduced-Order Modeling
- **Embedded code generation:** Simulink Coder, Embedded Coder
- **Rapid control prototyping & hardware-in-the-loop (HIL) testing:** NI VeriStand
- **Test and measurement system development:** NI LabVIEW, LabVIEW FPGA
- **Mechanical design (CAD):** CATIA, SolidWorks
- **Programming:** Python, C, MATLAB
- **Version control & management:** TortoiseSVN, JIRA

PUBLICATIONS AND PATENTS

- **Steering-Controlled Braking and Maneuvering System.** TR2025006494A2, 2025
- **Forced Response of Nonlinear Systems Under Multi-Frequency Excitations by Using Harmonic Balance Method.** *Nonlinear Structures & Systems, Vol. 1: Proceedings of the 42nd IMAC,* USA, 2025
- **Design and Verification of an Adaptive State-Tuned Power Management System for Series Hybrid Electric Tracked Vehicles.** *International Symposium on Advanced Vehicle Control,* Italy, 2024
- **Series-Hybrid Powertrains: Advancing Mobility Control in Electric Tracked Vehicle Technology.** *World Electric Vehicle Journal,* 2024
- **Adaptive Power Management System for Armored Series Hybrid Vehicles.** WO2024226022A1, 2024

LANGUAGES

Turkish (Native) English (Advanced) German (Basic)

ADDITIONAL INFORMATION

- **Interests:** Water Polo and Swimming (Professional), Gastronomy, Motorsports, Skiing
- **Driving License:** Class B
- **Military:** Completed